Project Report: Basic Calculator using Shall Script

BASIC CALCULATOR

Submitted by:

- **Mukul Dev Sankhla** — Roll No: 243501133 [B-1]

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Ashish Rai Sir, Department of Computer Applications

Project Guide / Mentor:
Ashish Rai Sir / Deepak Mathur Sir(mentor) Academic Year:

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Declaration

I hereby declare that the project work entitled "Basic Calculator" submitted to Lachoo Memorial College, BCA B1 IV Semester, is our own original work and has not been submitted earlier for award of any degree, diploma or certificate. (Signatures)

- Mukul Dev Sankhla — _____

Acknowledgement

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OBJECTIVES

The main objective of this project is to develop a simple calculator using Bash shell scripting in Linux.

This calculator is designed to perform basic arithmetic operations such as:

- Addition
- Subtraction
- Multiplication
- Division

The project also aims to provide a clear understanding of basic shell scripting concepts such as loops, conditional statements, user input handling, and arithmetic operations.

TOOLS AND TECHNOLOGIES USED

Operating System: Ubuntu Linux

Programming Language: Bash Shell Scripting

Interface Used: Terminal

PROJECT DESCRIPTION

This project is a menu-driven calculator program developed using shell scripting. The program allows the user to perform different arithmetic operations by selecting options from a menu.

At the start, the program displays a menu with different operations. The user is asked to enter two numbers and then choose an operation to perform.

Based on the user's choice, the program performs the selected operation and displays the result.

A loop is used to ensure that the program keeps running until the user selects the exit option. This makes the calculator interactive and user-friendly.

Special care has been taken in the division operation to handle division by zero. If the user enters zero as the second number, the program displays an error message instead of performing the operation.

ALGORITHM

Step 1: Start the program

Step 2: Display the calculator menu

Step 3: Take input for the first number

Step 4: Take input for the second number

Step 5: Ask the user to select an operation

Step 6: Use a case statement to determine the operation

Step 7: Perform the selected operation

Step 8: If division is selected, check if the second number is zero

Step 9: Display the result

Step 10: Ask the user to continue or exit

Step 11: Repeat the process until the user chooses to exit

Step 12: End the program

SOURCE CODE

```
#!/bin/bash

while true
do
echo "============================="
echo "     BASIC CALCULATOR"
echo "============================="

echo "Enter first number:"
read num1

echo "Enter second number:"
read num2
```

```
echo "Select operation:"
echo "1) Addition"
echo "2) Subtraction"
echo "3) Multiplication"
echo "4) Division"
echo "5) Exit"
read choice

case $choice in
  1)
    result=$((num1 + num2))
    echo "Result: $result"
    ;;
  2)
    result=$((num1 - num2))
    echo "Result: $result"
    ;;
  3)
    result=$((num1 * num2))
    echo "Result: $result"
    ;;
  4)
    if [ $num2 -eq 0 ]; then
      echo "Error: Cannot divide by zero!"
    else
      result=$(echo "scale=2; $num1 / $num2" | bc)
      echo "Result: $result"
    fi
    ;;
```

```
    5)
        echo "Exiting Calculator..."
        break
        ;;
    *)
        echo "Invalid choice! Try again."
        ;;
esac

echo ""
echo "Press Enter to continue..."
read

done
```

OUTPUT

The program displays a menu on the terminal and allows the user to enter two numbers. After selecting an operation, the result is displayed on the screen.

If the user selects division and enters zero as the second number, an error message is shown.

FUTURE ENHANCEMENTS

- Add more mathematical operations such as modulus and power
- Improve user interface

- Add input validation for better accuracy

- Convert the script into a graphical user interface (GUI) based application

CONCLUSION

This project helped in understanding the fundamentals of shell scripting in Linux. It provided hands-on experience with loops, conditional statements, and user input handling.

The calculator is simple, efficient, and demonstrates the practical use of scripting in solving basic computational problems.

THANK YOU!!